Discrete Probability Theory (IN0018)
School of Computation, Information, and Technology
Technical University of Munich

Tutorial 13

Confidence Interval (ci)

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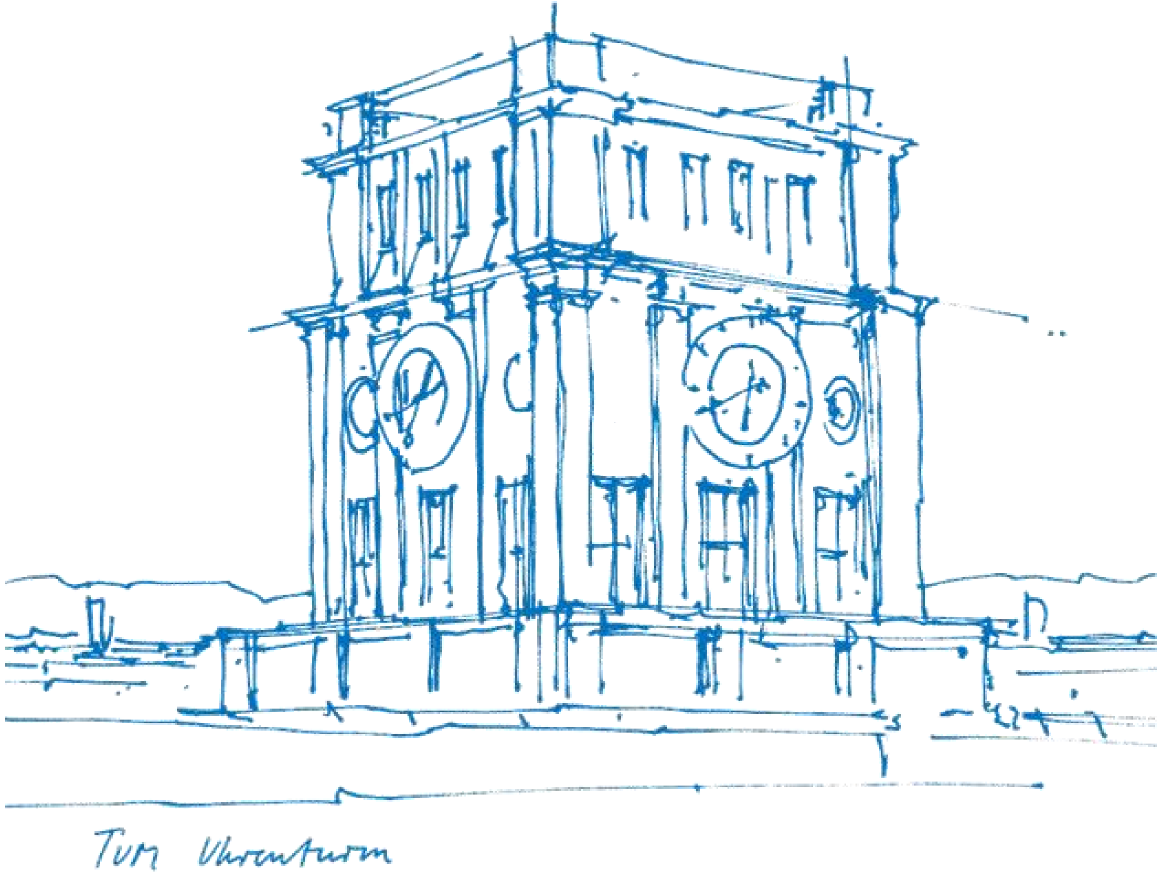

About Tutorial

- **Tutorial:** In-Person, 2h by two time periods per week - 12:00-14:00 Tue, 03.11.018 (Group B33)
  - 10:00-12:00 Fri, 01.06.011 (Group E24)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code ☞

https://drive.google.com/drive/folders1rBeL0NKHuYzrULK3Fr4CGr8svRhyTNOO?usp=sharing

➢ PART I. Recap of the Lecture

➢ PART II. Exercises from the Exercise Sheet

I. Recap

- **Confidence Interval**
  - ① - Sample Proportion
  - ② - Confidence Interval (ci) for a Proportion
    - - Standard Error (SE)
    - - Construction of Confidence Interval
  - ③ - Sample Size Determination by Margin of Error (ME)

Exercise 1 (confidence interval for a proportion)**.** A polling agency wants to estimate the proportion p of voters who support a particular candidate. They conduct a (uniformly!) random sample of $n = 400$ voters and find that $k = 220$ of them support the candidate.

(i) Calculate the sample proportion $\hat{p}$ of voters who support the candidate.

(ii) Construct an approximate 95% confidence interval for the true proportion p.

(iii) Interpret the confidence interval in the context of this problem.

(iv) Assume the agency wants to achieve a "margin of error" of ± 0.02 for the 95% ci. The margin of error is the quantity that gets added to and subtracted from the point estimate to obtain the ci. Approximately how many voters would they need to sample? (Assume $\hat{p}$ from the initial sample is a reasonable estimate for p for this planning.)

Q&A

$$P(\text{"You ace the exam"} \mid \text{"You paid attention and show up"}) = 1$$

—— Conditional probability tells you, showing up is already a win!

$$\mathbb{E}[\text{Your Score}] = \mathbb{E}[\text{Your Effort}] + \text{Random Luck}$$

—— May your effort be high and your luck always positive!

By the Law of Large Numbers:

$$\text{More practice} \Rightarrow \text{More stable performance}$$

—— Statistically speaking, you've got it!